

Rossmoyne Senior High School

Semester One Examination, 2018

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two:
Calculator-assumed

SOLUTIONS

Student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet.
3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.
4. Supplementary pages for the use of planning/continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
6. It is recommended that you do not use pencil, except in diagrams.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (98 Marks)

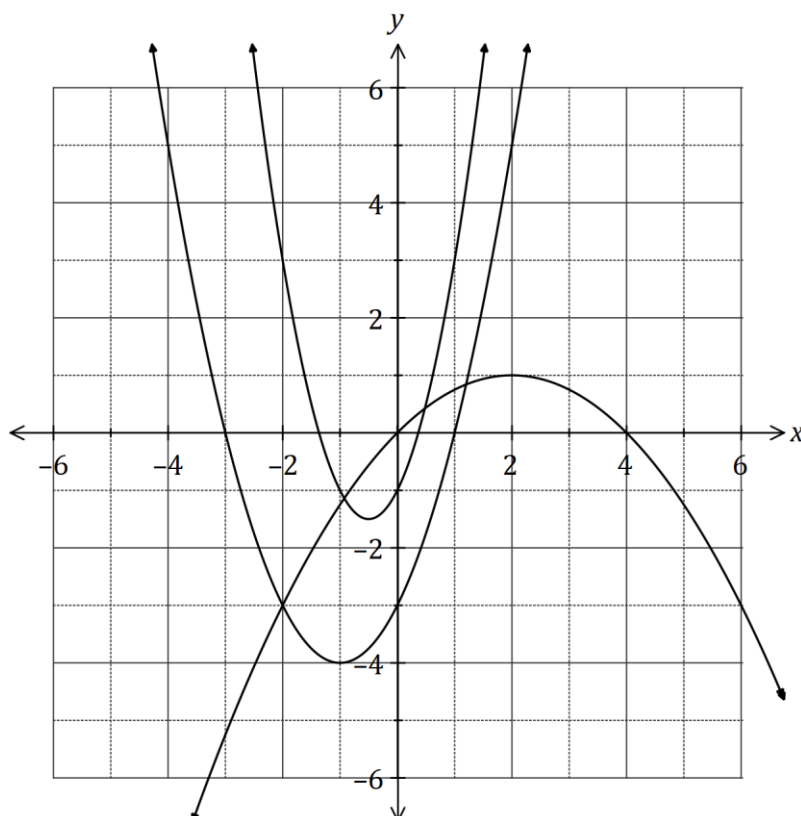
This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9

(4 marks)

The graphs of $y = 2x^2 + 2x + c$, $y = a(x - 2)^2 + 1$ and $y = (x + b)(x + 3)$ are shown below.



Determine the values of the constants a , b and c .

Solution
$x = 4 \Rightarrow 0 = a(4 - 2)^2 + 1$ $a = -\frac{1}{4}$
$b = -1 \text{ (Other root at -3)}$
$c = -1 \text{ (y-intercept)}$
Specific behaviours
✓ uses point on inverted parabola ✓ value of a ✓ value of b ✓ value of c

Question 10**(7 marks)**

An online store is offering new customers the opportunity to select 5 different products for just \$5 each. They can select from a range of 8 different books, 12 different magazines and 16 different games.

- (a) Determine how many different selections can be made.

(2 marks)

Solution
$8 + 12 + 16 = 36$ ${}^{36}C_5 = 376\,992$
Specific behaviours
✓ indicates choosing 5 from 36 ✓ correct number

- (b) Determine how many different selections can be made that just include games.

(2 marks)

Solution
${}^{16}C_5 = 4\,368$
Specific behaviours
✓ indicates choosing 5 from 16 ✓ correct number

- (c) Determine the probability that if a new customer randomly makes a selection, it will not include a book.

(3 marks)

Solution
$12 + 16 = 28$ ${}^{28}C_5 = 98\,280$ $P = \frac{98280}{376992} = \frac{195}{748} \approx 0.261$
Specific behaviours
✓ indicates choosing 8 from 26 ✓ forms probability ✓ simplifies or approximates with decimal

Question 11

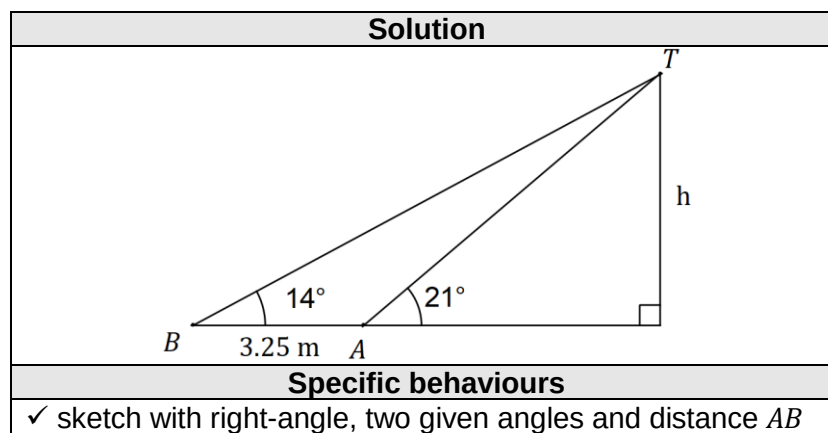
(6 marks)

A thin post stands vertically in the middle of a level playing ground. From point A on the ground, the angle of elevation to the top of the post, T , is 21° .

From point B , also on the ground but 3.25 metres further from the foot of the post than A , the angle of elevation to the top of the post is 14° .

(a) Draw a sketch to represent this information.

(1 mark)



(b) Showing use of trigonometry, determine the height of the post.

(5 marks)

Solution
$\frac{AT}{\sin 14} = \frac{3.255}{\sin 7} \Rightarrow AT = 6.45$ $h = 6.45 \times \sin 21 = 2.31 \text{ m}$
Specific behaviours
✓ angle BTA ✓ equation using sine rule ✓ solves for AT ✓ use of trig in right triangle ✓ determines h

Question 12

(6 marks)

- (a) Determine the equation of the axis of symmetry for the graph of $y = 4x^2 - 20x - 51$.

(2 marks)

Solution
$x = -\frac{b}{2a} = -\frac{-20}{2 \times 4} = 2.5$ $x = 2.5$
Specific behaviours
✓ indicates use of formula ✓ correct equation

- (b) The graph of $y = ax^2 + bx + 27$ passes through the points (3, 6) and (−4, −57). Determine the values of the constants a and b .

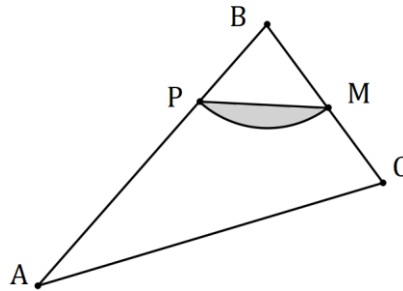
(4 marks)

Solution
$6 = (3)^2a + 3b + 27$ $6 = 9a + 3b + 27$ $-57 = (-4)^2a - 4b + 27$ $-57 = 16a - 4b + 27$ Solve simultaneously using CAS $a = -4, \quad b = 5$
Specific behaviours
✓ substitutes first point ✓ substitutes second point ✓ solves for a ✓ solves for b

Question 13

(10 marks)

A logo with triangular outline ABC contains a shaded segment bounded by the straight line PM and the circular arc PM with centre B and radius $BM = 25$ cm, as shown below.



Given that $\angle ABC = \frac{7\pi}{12}$, $\angle BCA = 4\angle BAC$ and M is the midpoint of BC , determine

- (a) the size of $\angle ABC$ in degrees.

(1 mark)

Solution
$\frac{7\pi}{12} \times \frac{180}{\pi} = 105^\circ$
Specific behaviours
✓ converts angle

- (b) the area of the shaded segment.

(2 marks)

Solution
$A = \frac{1}{2}(25)^2 \left(\frac{7\pi}{12} - \sin\left(\frac{7\pi}{12}\right) \right) \approx 270.8 \text{ cm}^2$
Specific behaviours
✓ indicates substitution into segment area formula
✓ evaluates area

- (c) the perimeter of the shaded segment.

(3 marks)

Solution
$PM_{arc} = 25 \times \frac{7\pi}{12} = \frac{175\pi}{12} \approx 45.81$
$b = \sqrt{25^2 + 25^2 - 2(25)(25)\cos 105} \approx 39.67$
Perimeter = $45.81 + 39.67 \approx 85.5 \text{ cm}$
Specific behaviours
✓ calculates arc length
✓ indicates use of cosine rule to find PM
✓ evaluates PM and states perimeter

(d) the area of triangle ABC .

(4 marks)

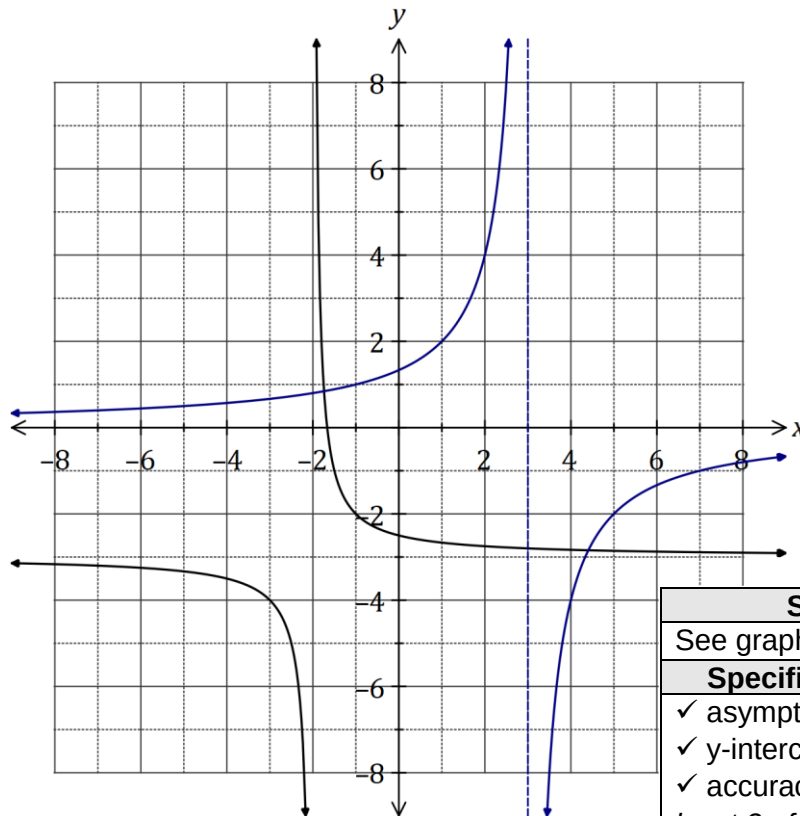
Solution
$\angle A + \angle C = 180 - 105$ $\angle A + 4\angle A = 105 \Rightarrow \angle A = 15$ $\frac{AC}{\sin 105} = \frac{2 \times 25}{\sin 15}$ $AC = 186.6$ $\text{Area} = \frac{1}{2}(50)(186.6) \sin(4 \times 15) \approx 4040 \text{ cm}^2$
Specific behaviours
✓ indicates use of equation to find second angle ✓ evaluates second angle and indicates use in sin rule ✓ evaluates second side ✓ evaluates triangle area

Question 14

(7 marks)

Let $f(x) = \frac{4}{3-x}$ and $g(x) = \frac{1}{x+p} + q$, where p and q are constants.

The graph of $y = g(x)$ is shown below.



Solution
See graph
Specific behaviours
✓ asymptotes ✓ y-intercept ✓ accuracy [i.e. <i>thu'</i> at least 3 of $(-1,1)$, $(1,2)$, $(2,4)$, $(4,-4)$ or $(7,-1)$]

(a) Sketch the graph of $y = f(x)$ on the axes above.

(3 marks)

(b) Determine the values of p and q .

(2 marks)

Solution
$p = 2, \quad q = -3$
Specific behaviours
✓ value of p ✓ value of q

(c) Solve the equation $f(x) = g(x)$, giving your solution(s) to one decimal place.

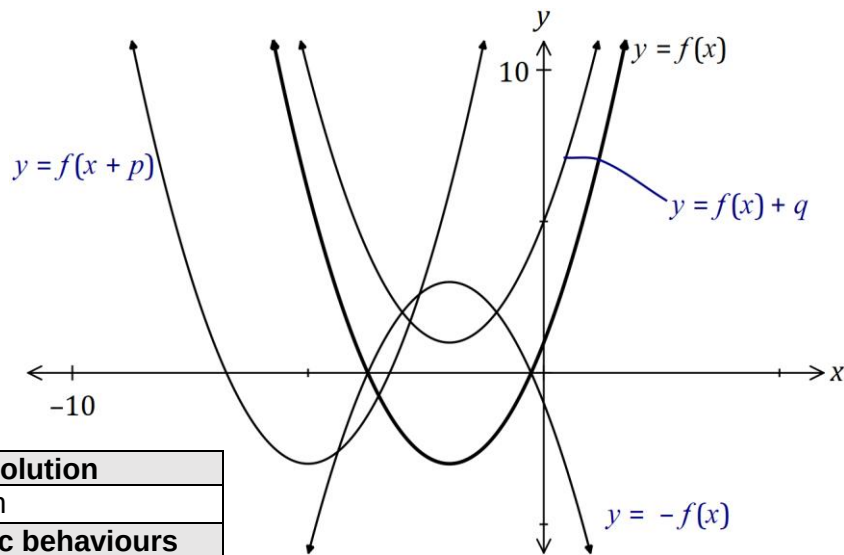
(2 marks)

Solution
$x = -1.7, \quad x = 4.4$
Specific behaviours
✓ one solution ✓ second solution <i>(Rounding for guidance only but penalise answers given as coordinates)</i>

Question 15

(6 marks)

- (a) The graph of $y = f(x)$ is shown in bold below. The graphs of $y = -f(x)$, $y = f(x + p)$ and $y = f(x) + q$ are also shown, where p and q are constants.



Solution
See graph
Specific behaviours
✓✓✓ each correct label

Clearly label the remaining graphs with $y = -f(x)$, $y = f(x + p)$ or $y = f(x) + q$.

(3 marks)

- (b) The one-to-one relation $y = 7 - 3x$ has domain and range given by $\{x: x = -2, 3, a\}$ and $\{y: y = -8, -2, b\}$ respectively. Determine the values of constants a and b . (3 marks)

Solution
$x = 3, y = -2$
$x = -2, y = 13 = b$
$x = a, y = 7 - 3a = -8 \Rightarrow a = 5$
Specific behaviours
✓ value of b
✓ indicates a is mapped onto -8
✓ solves for value of a

Question 16

(9 marks)

In a group of 250 students, it is known that 70% are in Year 12 and the rest in Year 11. 40% of those in Year 12 and 80% of those in Year 11 study maths, while the remainder do not.

Let T be the subset of students in Year 12 and M be the subset of students who study maths.

(a) Use set language and notation to describe the subset of students who

(i) do not study maths.

Solution
(i) $\bar{M}$
(ii) $M \cup T$
Specific behaviours
✓ correct notation
✓ correct notation
<i>(Penalise use of $P(\dots)$ once)</i>

(ii) study maths or are in Year 12.

(b) Determine the probability that a randomly chosen student from the group

(i) is in Year 11 and studies math.

Solution
$P(\bar{T} \cap M) = 0.3 \times 0.8 = 0.24 \quad \left(\equiv \frac{6}{25} \right)$
Specific behaviours
✓ correct probability

(ii) studies math.

Solution
$P(T \cap M) + P(\bar{T} \cap M) = 0.7 \times 0.4 + 0.24 = 0.28 + 0.24 = 0.52 \quad \left(\equiv \frac{13}{25} \right)$
Specific behaviours
✓ indicates suitable method
✓ correct probability

(iii) is in Year 11, given that they study math.

Solution
$P(\bar{T} M) = \frac{0.24}{0.52} = \frac{6}{13} \quad (\approx 0.4615)$
Specific behaviours
✓ uses (i) and (ii)
✓ expresses P correctly (no need to fully simplify)

(iv) is in Year 12 or studies math.

Solution
$P(T) + P(M) - P(T \cap M) = 0.7 + 0.52 - 0.28 = 0.94 \quad \left(\equiv \frac{47}{50} \right)$
Specific behaviours
✓ indicates suitable method
✓ correct probability

Question 17**(9 marks)**

For two events, X and Y , it is known that $P(X) = 0.48$ and $P(Y) = 0.35$. Determine the following probabilities, using any additional information only within that part of the question.

(a) $P(\bar{Y})$.

(1 mark)

Solution
$P(\bar{Y}) = 1 - 0.35 = 0.65$
Specific behaviours
✓ correct probability

(b) $P(Y \cap \bar{X})$ when $P(X \cap \bar{Y}) = 0.29$.

(2 marks)

Solution
$P(X \cap Y) = 0.48 - 0.29 = 0.19$
$P(Y \cap \bar{X}) = 0.35 - 0.19 = 0.16$
Specific behaviours
✓ calculates $P(X \cap Y)$
✓ correct probability

(c) $P(X|Y)$ when $P(Y|X) = 0.7$.

(3 marks)

Solution
$P(X \cap Y) = 0.48 \times 0.7 = 0.336$
$P(X Y) = \frac{0.336}{0.35} = 0.96$
Specific behaviours
✓ indicates use of conditional probability rule
✓ calculates $P(X \cap Y)$
✓ correct probability

(d) $P(X \cup Y)$ when X and Y are independent.

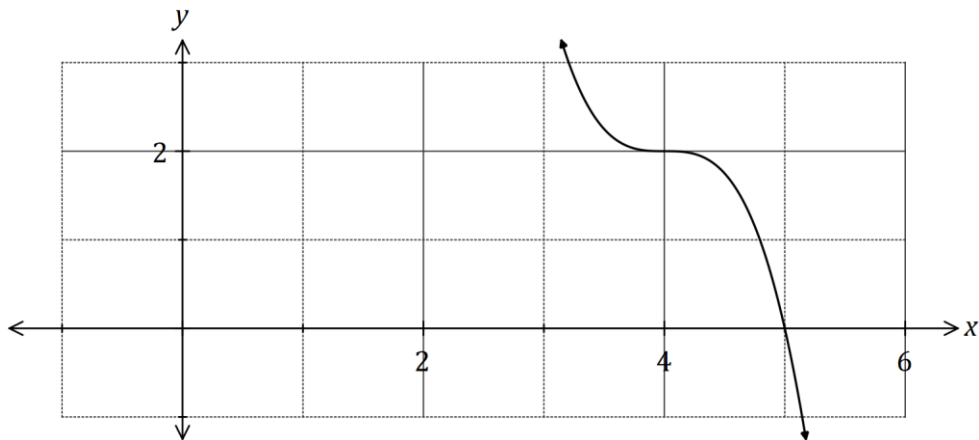
(3 marks)

Solution
$P(X \cap Y) = 0.48 \times 0.35 = 0.168$
$P(X \cup Y) = 0.48 + 0.35 - 0.168$ $= 0.662$
Specific behaviours
✓ uses independence to calculate $P(X \cap Y)$
✓ indicates use of addition rule
✓ correct probability

Question 18

(6 marks)

- (a) Part of the graph of $y = f(x)$ is shown below, where $f(x) = -2(x - b)^3 + c$, and b and c are constants.



- (i) State the degree of $f(x)$.

Solution
3
Specific behaviours
✓ correct degree

(1 mark)

- (ii) Determine the value of b .

Solution
$b = 4$
Specific behaviours
✓ correct value

(1 mark)

- (iii) Determine $f(0)$.

Solution
$f(x) = -2(x - 4)^3 + 2$
$f(0) = -2(-4)^3 + 2 = 130$
Specific behaviours
✓ indicates value of c
✓ evaluates

(2 marks)

- (b) Another function is given by $g(x) = f(x + 8)$.

Describe how to obtain the graph of $y = g(x)$ from the graph of $y = f(x)$.

(2 marks)

Solution
Translate graph 8 units to the left.
Specific behaviours
✓ uses translation
✓ indicates distance and direction

Question 19

(12 marks)

During 2018, the altitude of the sun, θ degrees, at noon in Melbourne on the n^{th} day of the year can be modelled by the equation

$$\theta = 23.5 \sin\left(\frac{8\pi(101 + n)}{1461}\right) + 52.2$$

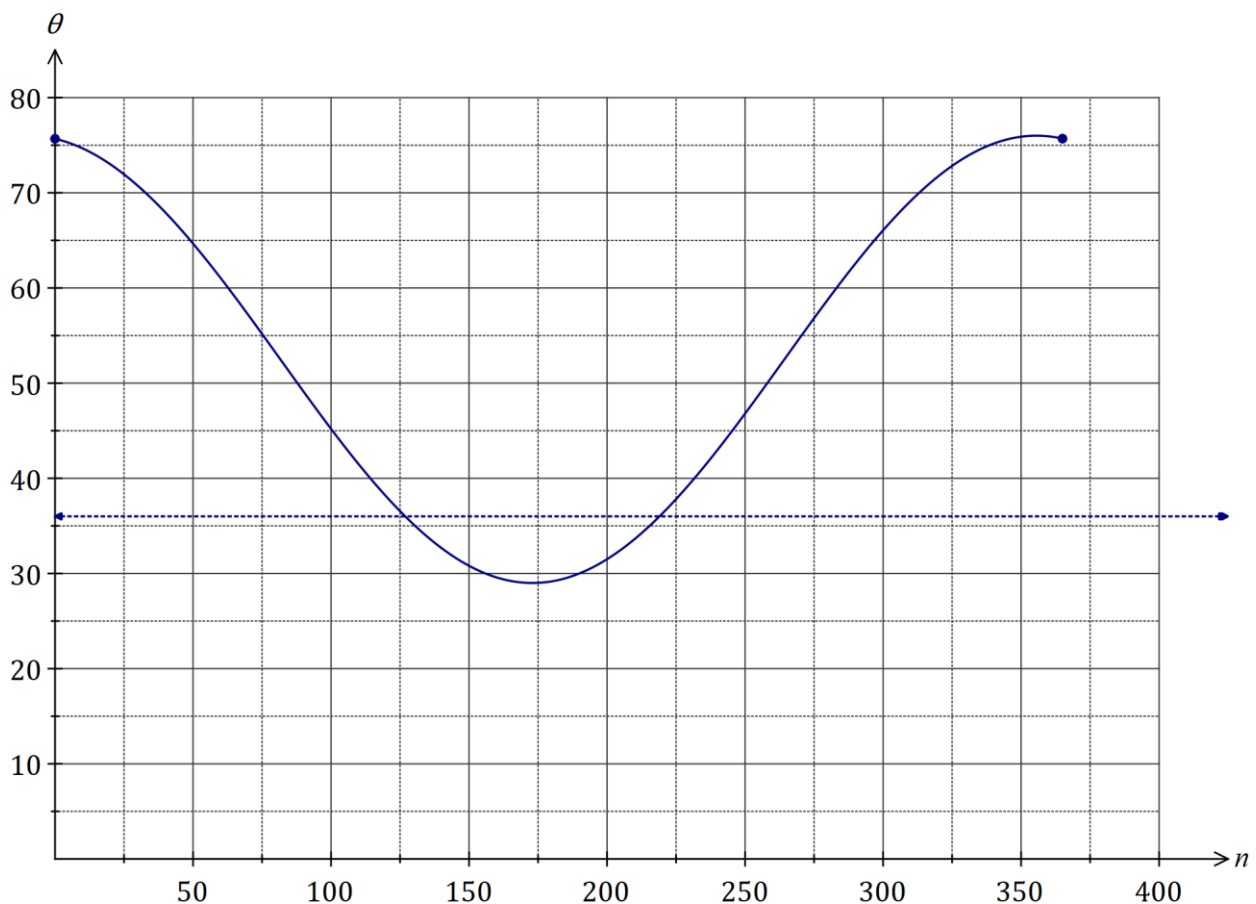
- (a) On the 26th of January, the altitude was 71.4° . Calculate the altitude ten days earlier.

(2 marks)

Solution	
$n = 16,$	$\theta = 73.4^\circ$
Specific behaviours	
✓ indicates n	
✓ correct angle	

- (b) Graph the altitude on the axes below for $0 \leq n \leq 365$.

(4 marks)



Solution	
See graph	
Specific behaviours	
✓ endpoints, $\theta \approx 75^\circ$	
✓ minimum close to $(173, 29^\circ)$	
✓ maximum close to right endpoint	
✓ smooth curve	

See next page

- (c) State the minimum altitude of the sun at noon in Melbourne and on which day of the year this occurred. (2 marks)

Solution
$\theta_{MIN} = 28.7^\circ$ on day 173
Specific behaviours
✓ altitude, correct to 1dp ✓ day of year, rounded to whole number

- (d) Solar panels on the roof of a Melbourne business are designed to meet its entire power needs on cloudless days when the altitude of the sun is at least 36° at noon.

- (i) Draw a straight line on your graph to represent this requirement. (1 mark)

Solution
See graph - horizontal line $\theta = 36^\circ$
Specific behaviours
✓ neat, straight line

- (ii) Determine the number of days the panels are expected to achieve this aim during 2018, ignoring the possibility of cloud cover. (3 marks)

Solution
$\theta > 36^\circ \Rightarrow 126 \leq n \leq 220$ $220 - 126 = 94$ $365 - 94 = 271 \text{ days}$
Specific behaviours
✓ one value of n ✓ second value of n ✓ maximum number of days

Question 20

(9 marks)

Injury events in a state were grouped as follows: (D) injury events resulting in death; (H) injury events requiring hospitalisation; and (E) injury events requiring treatment at an emergency department. Injury events were further grouped according to whether prior use of alcohol was a contributing factor. The number of injury events for one month is summarised in the table below.

	D	H	E	Total
Alcohol	25	581	5 605	6 211
No Alcohol	115	4 422	11 912	16 449
Total	140	5 003	17 517	22 660

- (a) Calculate the percentage of all injury events that involved prior use of alcohol. (2 marks)

Solution
$\frac{6211}{22660} \times 100 = 27.4\%$
Specific behaviours
✓ identifies correct totals ✓ correct percentage

- (b) If one injury event was selected at random from the data, determine the probability that

- (i) it required hospitalisation.

Solution
$P(H) = \frac{5003}{22660} \approx 0.221$
Specific behaviours
✓ correct probability

(1 mark)

- (ii) it did not require treatment at an emergency department given that prior use of alcohol was a contributing factor. (2 marks)

Solution
$P(E A) = \frac{25 + 581}{6211} = \frac{606}{6211} \approx 0.0976$
Specific behaviours
✓ numerator ✓ denominator

- (iii) prior use of alcohol was a contributing factor given that it required hospitalisation. (2 marks)

Solution
$P(A H) = \frac{581}{5003} \approx 0.116$
Specific behaviours
✓ numerator ✓ denominator

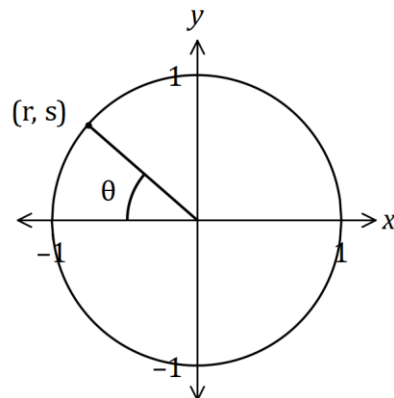
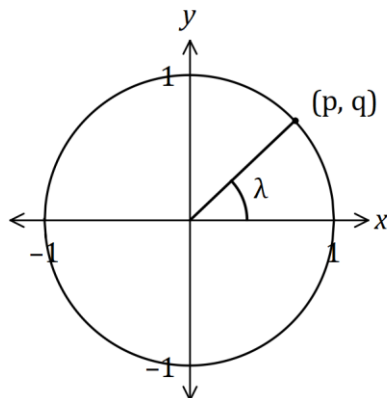
- (c) Use your answers above to explain whether the type of injury event (D, H or E) appears to be independent of prior use of alcohol. (2 marks)

Solution
From (a), $P(A) = 0.274$ but from (b)(iii) $P(A H) = 0.116$ and since $P(A) \neq P(A H)$ then events NOT independent.
Specific behaviours
✓ states not independent ✓ explains using existing answers

Question 21

(7 marks)

Consider the points with coordinates (p, q) and (r, s) that lie in the first and second quadrants respectively of the unit circles shown below, where λ and θ are acute angles.



Determine the following in terms of p, q, r and s , simplifying your answers where possible.

(a) $\sin \theta$.

(1 mark)

(b) $\tan \theta$.

(1 mark)

(c) $\cos(\pi + \lambda)$.

(1 mark)

(d) $\sin(90^\circ - \lambda)$.

(1 mark)

Solutions	
(i) s	
(ii) $\frac{s}{r}$	
(iii) $-p$	
(iv) p	
Specific behaviours	
✓ each correct response	

(e) $\cos(\theta + \lambda)$.

(3 marks)

Solution
$\begin{aligned}\cos(\theta + \lambda) &= \cos \theta \cos \lambda - \sin \theta \sin \lambda \\ &= (-r)(p) - (s)(q) \\ &= -pr - qs\end{aligned}$
Specific behaviours
✓ uses identity ✓ at least two correct trig values ✓ correct expression

Supplementary page

Question number: _____

Supplementary page

Question number: _____

